Week 15 Homework

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In order to receive full credit problems and solutions must be clearly written and presented neatly. In writing up your solution, you should state the problem as explaining the solution. Merely giving answers to exercises will is not enough to receive credit. Your write up must show work and convince me that you understand the material. If you need additional time to complete this assignment, please email me. Starred () proof problems are eligible for inclusion in the proof portfolio assignment.

Exercises

Exercise

In this problem, boxes can be empty. How many ways are there to distribute 6 balls in to 3 boxes if

1. both the balls and boxes are labeled?
2. balls are labeled, but the boxes are unlabeled?
3. the balls are unlabeled, but the boxes are labeled?
4. both the boxes and balls are unlabeled?

Exercise

In this problem, boxes must be non-empty. How many ways are there to distribute 8 balls in to 4 boxes if

1. both the balls and boxes are labeled?
2. balls are labeled, but the boxes are unlabeled?
3. the balls are unlabeled, but the boxes are labeled?
4. both the boxes and balls are unlabeled?

Exercise

How many 8-bit strings begin with a 010 or begin with 11? How many 8-bit strings begin with a 010 or end with 11?

Exercise

Find the number of solutions to $x + y + z = 100$ in non-negative integer x, y, and z, where $x \leq 50$, $y \leq 40$, and $z \leq 30$.

Exercise

How many permutations of the numbers 1 through 8 are there in which none of the terms 12, 34, 56, 78 appear?

Exercise

How many permutations of the numbers 1 through 8 are there in which none of the even integers remain in its natural position?

Exercise

Do the enigma machine example from class where the number of rotors is 8, rather than 5.

Exercise

How many permutations of 1,1,2,2,3,3,4,4,5,5 are there in which no two adjacent numbers are equal?

Exercise

Suppose you roll a 6 sided die 12 times. In how many ways can all 6 six numbers appear?

Exercise

How many integers in the set of integers from 1 to 1000 are divisible by a prime number less than 10?

Exercise

A bridge hand consists of 13 cards from a standard 52 card deck. How many different hands contain at least on card from each of the four suits?

Exercise

Let $A = \{1, 2, 3, 4, 5\}$. How many injective functions $f : A \to A$ have the property that for each x in A, $f(x) \neq x$?

Exercise

Suppose you planned on giving 7 gold stars to some of the 13 star students in your class. Each student can receive at most one star. How many ways can you do this?

Exercise

What is the formula for the number of ways to place n unlabeled balls into k unlabeled boxes **without** the restriction that no box may be empty?

Exercise

How many ways are there to place $n + k$ stars on a $2 \times n$ (two rows, n columns) board with no columns having no stars? Here $k \geq 0$.

Exercise

What is the formula for the number of ways to place n labeled balls into k unlabeled boxes **without** the restriction that no box may be empty?

Proofs

Proof

Prove by induction $S(n, 3) > 3^{n-2}$ for all $n \geq 6$.

Proof

Prove

$(1-a_1)(1-a_2)\cdots(1-a_k) = 1-(a_1+a_2+\cdots+a_k)+(a_1a_2+\cdots+a_{k-1}a_k)+\cdots+(-1)^k(a_1\cdots a_k)$

Proof

For fixed m, find and prove the number of Hamiltonian Paths on on $K_{m,n}$ as a function of n.

Proof

From the text, Exercise 4.6

Proof

From the text, Exercise 4.7

Proof

Let S(n,k) be the Stirling numbers of the second kind. Prove
a. $S(n, n-1) = \binom{n}{2}$
b. $S(n, n-2) = \binom{n}{3} + 3\binom{n}{4}$

Proof

Let S(n,k) be the Stirling numbers of the second kind. Prove $S(n,k) = \sum_{m=k-1}^{n-1} \binom{n-1}{m} S(m, k-1)$

Proof

Show that if n is a positive integer, then

$$n! = \binom{n}{0} d_n + \binom{n}{1} d_{n-1} + \binom{n}{2} d_{n-2} + \cdots + \binom{n}{n-1} d_1 + \binom{n}{n} d_0$$

where $d_n = (n-1)(d_{n-1} + d_{n-2})$ with $d_1 = 0$ and $d_2 = 1$ is the number of derangements of n objects.

Proof

Let A be a subset of B with $|A| = m$ and $|B| = n$, and let r be an integer $m \leq r \leq n$. Show the number of r element subsets of B which contain A as a subset is $\binom{n-m}{r-m}$.

Proof

Use the Inclusion-Exclusion Principle to show that for any non-negative integers m, r, n such that $m \leq r \leq n$,

$$\binom{n-m}{r-m} = \sum_{i=0}^{m} (-1)^i \binom{m}{i} \binom{n-i}{r}.$$